

Linear Regression Model: House Price Predictor

AI/ML Internship — Task 1 | California Housing Dataset

1. Objective

This project applies a Linear Regression model to predict median house values in California districts using the California Housing dataset. The goal is to practice the complete machine learning workflow: data loading, exploratory data analysis (EDA), preprocessing, model training, evaluation, and reporting.

2. Dataset Overview

The dataset contains 20,640 records with 8 numeric features (median income, house age, average rooms, average bedrooms, population, average occupancy, latitude, and longitude) and one target variable, median house value (in \$100,000s). No missing values were found in any column.

3. Exploratory Data Analysis

Summary statistics revealed several noteworthy patterns. Median income (MedInc) showed the strongest correlation with house value ($r = 0.69$), while most other features showed weak correlations (below 0.15 in magnitude). Two outlier-prone columns were identified: AveRooms (max value of 141.9 vs. a mean of 5.4) and AveOccup (max value of 1,243 vs. a mean of 3.1), both likely reflecting a small number of unusual districts rather than data errors. Additionally, the target variable MedHouseVal is capped at 5.00001 (equivalent to \$500,000), meaning any district with a true median value above this threshold is recorded at the cap. This is a known limitation of the dataset and affects both model training and evaluation, as discussed below.

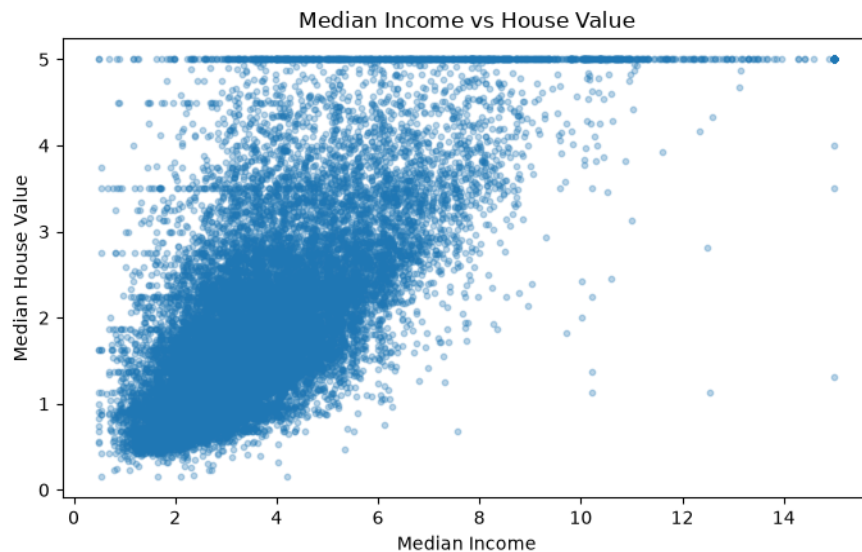


Figure 1: Median income shows a clear positive relationship with house value. The horizontal band of points at value = 5.0 reflects the dataset's price cap.

4. Methodology

All 8 available features were used as predictors. The data was split into training (80%, 16,512 rows) and test (20%, 4,128 rows) sets using a fixed random seed for reproducibility. A standard Linear Regression model (scikit-learn) was trained on the training set and evaluated on the held-out test set.

5. Results

Metric	Value	Interpretation
MAE	0.533	Average prediction error of ~\$53,300
RMSE	0.746	~\$74,600, more sensitive to large errors
R ² Score	0.576	Model explains ~57.6% of price variation

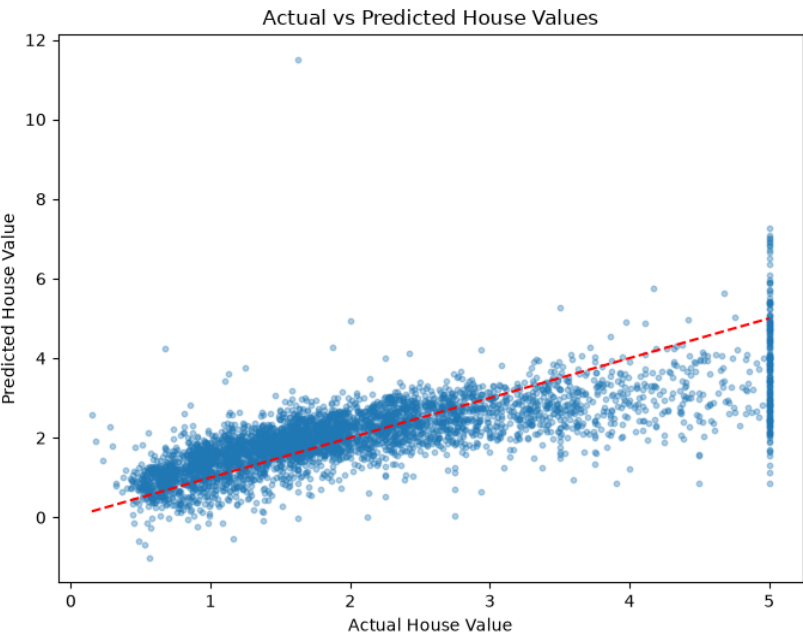


Figure 2: Predicted vs. actual house values. Most points cluster near the red reference line, indicating reasonable predictive performance. A vertical smear near actual = 5.0 corresponds to price-capped districts, and one notable outlier was predicted far above its actual value.

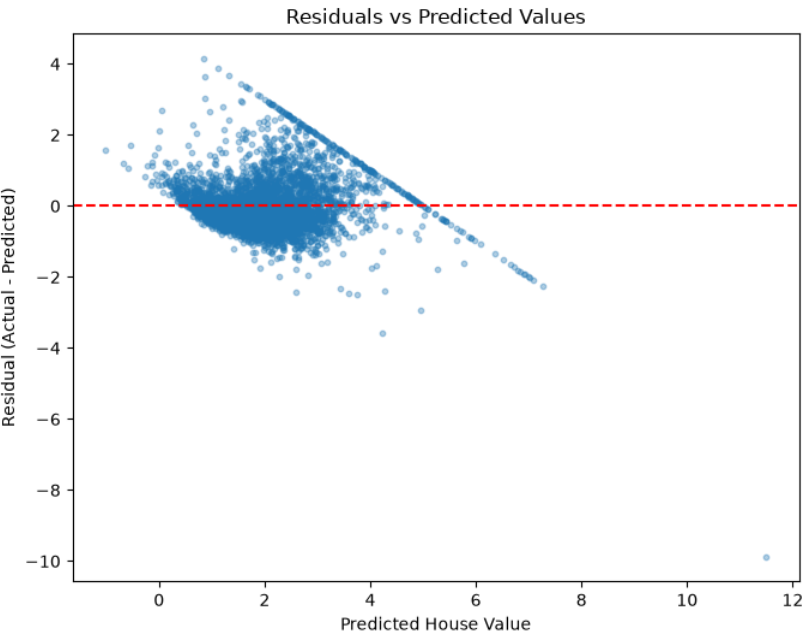


Figure 3: Residuals vs. predicted values. Errors are broadly centered around zero, with a distinct diagonal band caused by the price cap, and mild widening of spread at higher predicted values.

6. Limitations & Improvement Ideas

- The \$500,000 price cap censors true values for expensive districts, introducing systematic error that a linear model cannot resolve.
- Linear Regression assumes straight-line relationships; the data likely contains non-linear patterns (e.g., location-based pricing effects) that a more flexible model could capture.
- Trying ensemble models such as Random Forest or Gradient Boosting Regressor would likely improve R^2 , as these can model non-linear interactions between features.
- Feature engineering, such as creating a rooms-per-household or bedrooms-per-room ratio, could improve signal beyond the raw features used here.
- Removing or capping extreme outliers in AveRooms and AveOccup could reduce their disproportionate influence on the fitted line.

7. Conclusion

The Linear Regression model provides a reasonable baseline for predicting California house prices, explaining approximately 58% of the variance in the test set with a typical error of roughly \$53,000–\$75,000. Median income emerged as the dominant predictor. While adequate as an introductory model, the results point clearly toward more flexible, non-linear models as a next step for improved accuracy.